Segmentation Method Comparison for AAPS-Guided Organoid Quantification

Purpose

This document records the segmentation comparison used to justify the final AAPS method routing for the App80, App65, and App81 organoid image datasets. The comparison is intentionally based on representative source-matched fields from the historical runs rather than on re-rendered schematic examples. The figures are supplied without embedded text so that labels can be added or adjusted later in vector-layout software. The full row and column definitions are therefore given in the captions and in the source manifest `segmentation_method_comparison_sources.json`.

Comparison Strategy

The comparison follows the same logic used during the AAPS workflow. First, a simple or early segmentation attempt was inspected for each biological imaging setting. Second, failure modes were identified at the image level: missing large organoids, over-segmentation of debris or hydrogel boundaries, fragmented masks, or failure to separate adjacent instances. Third, the method was routed to a more appropriate segmentation branch using image priors such as organoid size, contrast, density, modality, and the expected biological endpoint.

For App80 Y-27632 recovery images, early watershed and SAM-style proposals were compared with the final multiscale large-object recovery workflow. For App65 alginate images, old multiscale and threshold-hinted Cellpose outputs were compared with the final direct-brightfield workflow. For App81 density images, threshold-hinted Cellpose, resized multiscale Cellpose, image-generation-assisted masks, and the final stitched hybrid outputs were compared on the same low density examples.

Observed Failure Modes and Routing Decisions

Figure Captions

Manuscript Interpretation

These comparisons support the manuscript claim that the AAPS harness does not depend on one universal segmentation algorithm. Instead, it records dataset context, tests candidate segmentation branches, preserves intermediate outputs, and routes each biological experiment to the method that best supports the downstream measurement. This is important for organoid assays because growth, differentiation, and fusion are not all optimized by the same mask behavior: growth metrics favor robust area recovery, differentiation metrics depend on darkness or wall-intensity stability, and fusion metrics require clean instance boundaries and spatial structure.

The final figures should be cited as method-comparison evidence, not as a standalone quantitative benchmark. They demonstrate visually why the final App80, App65, and App81 pipelines were selected

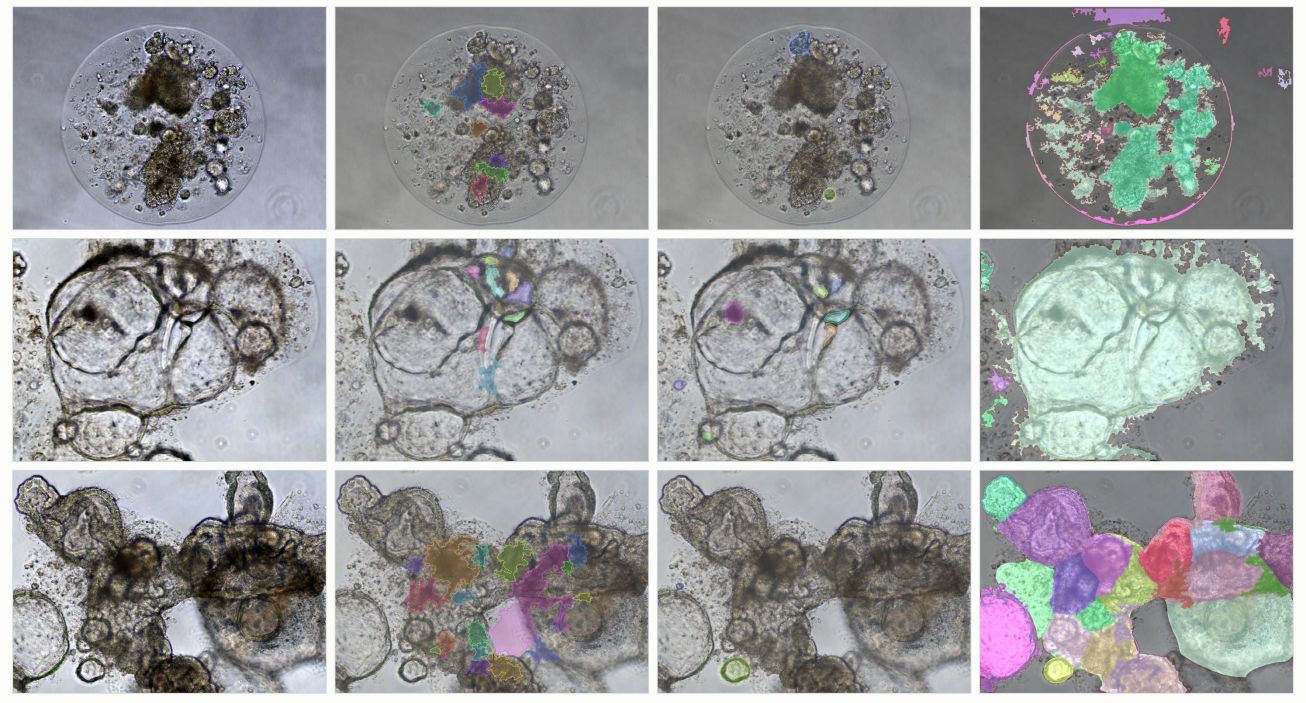


Figure 1: **App80 Y-27632 segmentation comparison.** Rows show representative 10 uM Y-27632 fields from 5, 10, and 12 December 2025, respectively. Columns show, from left to right: the raw brightfield image, the early prompted-watershed overlay, the early CPSAM/SAM-style proposal overlay, and the final AAPS multiscale large-object recovery overlay. The comparison illustrates why a plain edge- or prompt-driven proposal was insufficient for the App80 recovery experiment: organoid size and contrast changed over time, and large or weakly bounded structures required a multiscale branch rather than a single local-threshold or proposal method.

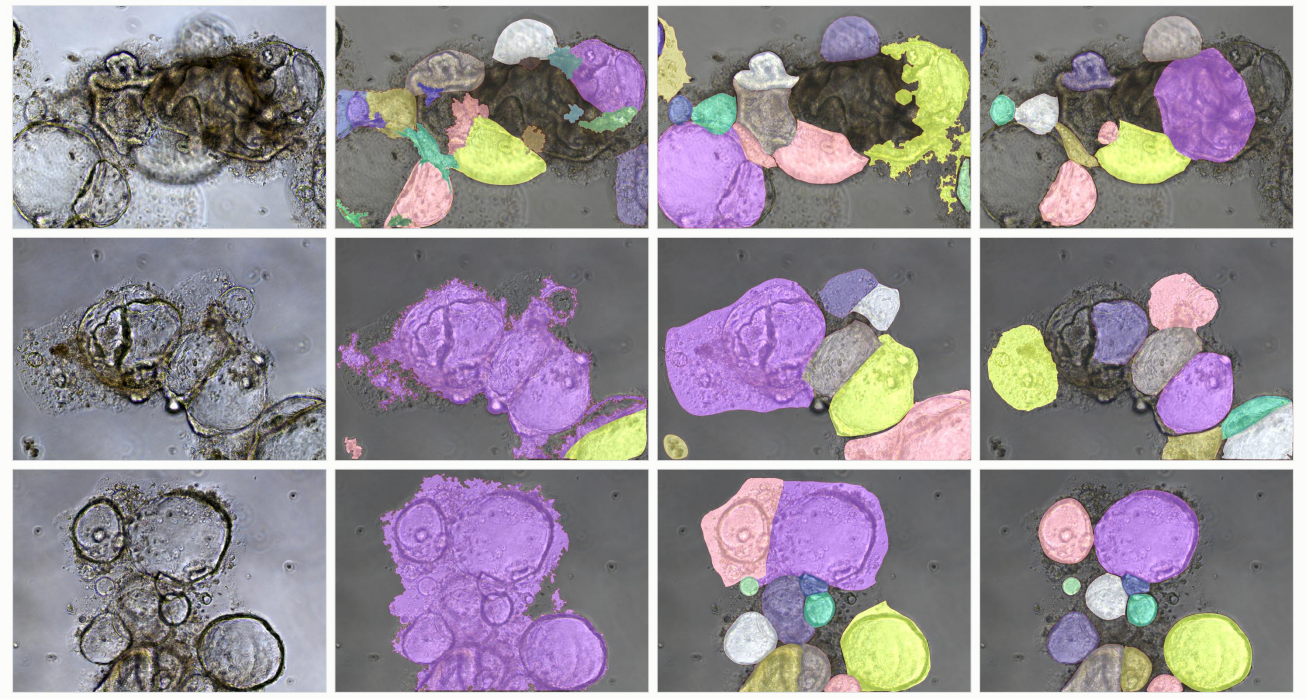


Figure 2: **App65 alginate segmentation comparison.** Rows show representative 0.02% alginate fields from 29 November 2025 (10x00, 10x14, and 10x15). Columns show, from left to right: the raw brightfield image, an older multiscale Cellpose large-recovery overlay, a threshold-hinted Cellpose test overlay, and the final AAPS direct-brightfield segmentation overlay. The comparison emphasizes that alginate-induced morphology requires a branch that follows the actual brightfield organoid wall and internal contrast, rather than over-weighting generic threshold evidence or a single diameter prior.

Table 1: Segmentation failure modes motivating AAPS method routing.

Dataset	Early-method limitations	Final routing rationale
App80 Y-27632	Prompted watershed and SAM-style proposals were sensitive to edge contrast and local debris. They could split continuous organoids, miss large low-contrast bodies, or return proposals that did not correspond to analyzable organoid instances.	The final multiscale large-recovery branch emphasized organoid-scale connected components and recovered larger objects that were biologically relevant for growth, average-area, darkness, and fusion metrics.
App65 alginate	Old multiscale and threshold-hinted Cellpose outputs improved over plain thresholding but were still unstable when wall contrast, internal darkness, and alginate-dependent morphology changed across images.	The final direct-brightfield branch used the organoid boundary and wall signal more directly, producing cleaner instances for perimeter, wall darkness, and peripheral/central fusion-related metrics.
App81 density	A single Cellpose diameter or threshold hint did not cover the full range of cystic organoid sizes. Image-generation masks were useful for identifying cystic structures but could merge adjacent instances or hallucinate smooth droplet boundaries without QC.	A stitched hybrid workflow allowed method selection by panel and replicate: Cellpose diameter variants handled most fields, while image-generation-assisted rescue was used only where it improved missed cystic objects after human-in-the-loop QC.

and why difficult panels were handled by AAPS-style quality control and human-in-the-loop refinement rather than by unguarded autonomous segmentation.

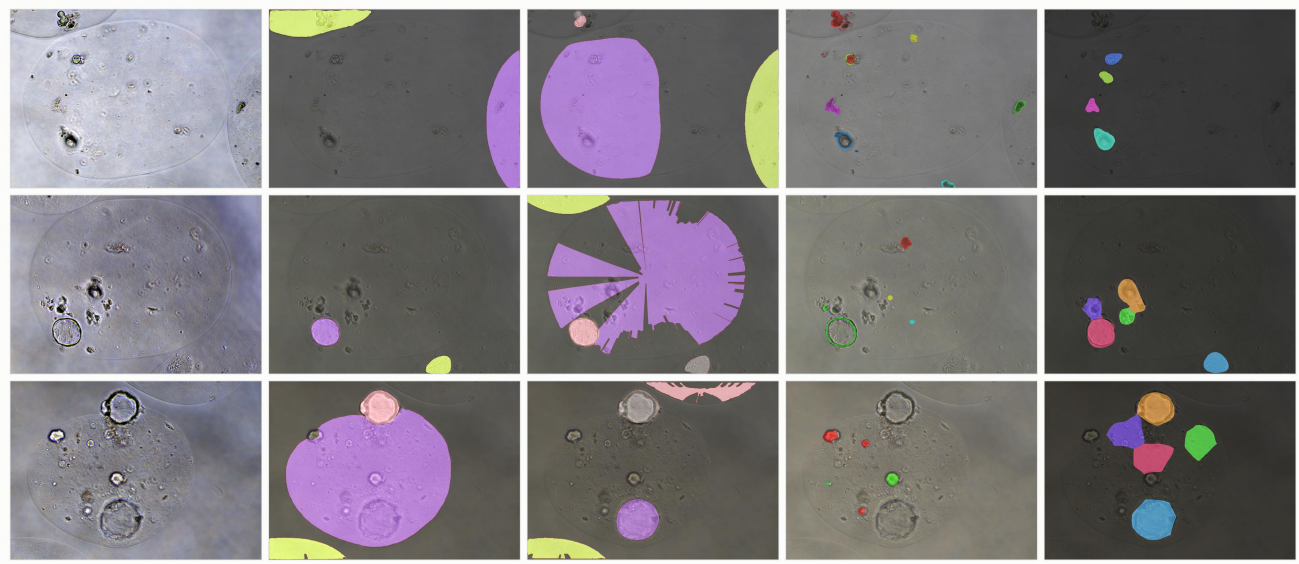


Figure 3: **App81 density segmentation and image-generation-assisted method routing.** Rows show low-density App81 examples from 14, 15, and 16 October 2025, respectively. Columns show, from left to right: the raw brightfield image, threshold-hinted Cellpose, resized multiscale Cellpose, the image-generation masking result, and the final stitched AAPS-selected hybrid overlay. The figure summarizes the main App81 lesson: threshold and Cellpose variants are efficient and reproducible for many panels, while image generation is useful as a targeted semantic prior for cystic organoids. However, image-generation outputs must be harnessed by panel-level QC and instance cleanup because they may merge adjacent organoids or confuse a pale hydrogel boundary with an organoid.